

Derivation v19: From 5D Action to 4D Newton Law and Calibrated Closure

EDC Collaboration

Derivation note; explicit steps shown

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Abstract

This note presents a self-contained derivation of the bridge relation $M_{\text{Pl}}^2 = M_5^3 \mathcal{I}$ from the 5D Einstein-Hilbert action, followed by explicit evaluation of the normalization integral $\mathcal{I}$ for the compact flat extra dimension. The calibrated closure to the 5D Planck mass M_5 is then presented with full epistemic tagging. All steps are shown explicitly; this is a derivation document, not a summary.

1 Setup: 5D Geometry and Action

1.1 Manifold Structure

We consider a 5-dimensional spacetime $\mathcal{M}^5$ with coordinates $x^A = (x^\mu, \xi)$ where $\mu = 0, 1, 2, 3$ labels the 4D brane directions and ξ is the extra-dimensional coordinate. The 4D brane Σ^4 is located at $\xi = 0$.

The 5D metric takes the warped form:

$$ds^2 = g_{AB} dx^A dx^B = e^{2A(\xi)} \eta_{\mu\nu} dx^\mu dx^\nu + d\xi^2 \quad (1)$$

where $A(\xi)$ is the warp factor and $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ is the 4D Minkowski metric.

1.2 Bulk Action

The gravitational dynamics are governed by the 5D Einstein-Hilbert action:

$$S_{\text{bulk}} = \frac{1}{2\kappa_5^2} \int_{\mathcal{M}^5} d^5x \sqrt{-g^{(5)}} \left(R^{(5)} - 2\Lambda_5 \right) \quad (2)$$

where $\kappa_5^2 = 8\pi/M_5^3$ is the 5D gravitational coupling, $R^{(5)}$ is the 5D Ricci scalar, and Λ_5 is the bulk cosmological constant.

The Gibbons-Hawking-York boundary term ensures a well-posed variational problem:

$$S_{\text{GHY}} = \frac{1}{\kappa_5^2} \int_{\partial\mathcal{M}} d^4x \sqrt{-\gamma} K \quad (3)$$

where $\gamma_{\mu\nu}$ is the induced metric on the boundary and K is the trace of the extrinsic curvature.

1.3 Brane Action

The brane at $\xi = 0$ carries tension σ :

$$S_{\text{brane}} = -\sigma \int_{\Sigma^4} d^4x \sqrt{-\gamma} \quad (4)$$

The Israel junction conditions at the brane location relate the jump in extrinsic curvature to the brane stress-energy:

$$[K_{\mu\nu}] - \gamma_{\mu\nu}[K] = -\kappa_5^2 \left(T_{\mu\nu}^{\text{brane}} - \frac{1}{3}\gamma_{\mu\nu}T^{\text{brane}} \right) \quad (5)$$

For a pure-tension brane, this yields:

$$[K_{\mu\nu}] = -\frac{\kappa_5^2\sigma}{3}\gamma_{\mu\nu} \quad (6)$$

2 Linearized Gravity and Mode Decomposition

2.1 Metric Perturbation

We perturb around the background (1):

$$g_{AB} = \bar{g}_{AB} + h_{AB} \quad (7)$$

where $\bar{g}_{AB}$ is the background metric and h_{AB} is the perturbation.

For the graviton sector, we focus on the transverse-traceless (TT) components $h_{\mu\nu}^{\text{TT}}$ satisfying:

$$\partial^\mu h_{\mu\nu}^{\text{TT}} = 0, \quad \eta^{\mu\nu} h_{\mu\nu}^{\text{TT}} = 0 \quad (8)$$

2.2 Kaluza-Klein Decomposition

We decompose the 4D graviton field into Kaluza-Klein modes:

$$h_{\mu\nu}^{\text{TT}}(x, \xi) = \sum_n h_{\mu\nu}^{(n)}(x)\psi_n(\xi) \quad (9)$$

where $h_{\mu\nu}^{(n)}(x)$ are the 4D mass eigenstates and $\psi_n(\xi)$ are the extra-dimensional wavefunctions.

2.3 Equation of Motion for Mode Functions

Substituting the decomposition (9) into the linearized 5D Einstein equation:

$$\square^{(5)} h_{\mu\nu}^{\text{TT}} = 0 \quad (10)$$

and using the warped metric (1), we obtain:

$$e^{-2A}\square^{(4)}h_{\mu\nu}^{(n)}\psi_n + h_{\mu\nu}^{(n)}[\psi_n'' + 4A'\psi_n'] = 0 \quad (11)$$

where primes denote ∂_ξ and $\square^{(4)} = \eta^{\mu\nu}\partial_\mu\partial_\nu$.

Separating variables with $\square^{(4)}h_{\mu\nu}^{(n)} = -m_n^2h_{\mu\nu}^{(n)}$:

$$\psi_n'' + 4A'\psi_n' + m_n^2e^{2A}\psi_n = 0 \quad (12)$$

2.4 Zero-Mode Solution

For the massless graviton ($m_0 = 0$), equation (12) becomes:

$$\psi_0'' + 4A'\psi_0' = 0 \quad (13)$$

This can be rewritten as:

$$\frac{d}{d\xi}(e^{4A}\psi_0') = 0 \quad (14)$$

Integrating once:

$$e^{4A}\psi_0' = C_1 = \text{const} \quad (15)$$

For a normalizable zero-mode localized near the brane, we require $C_1 = 0$, giving:

$$\psi_0' = 0 \quad \Rightarrow \quad \psi_0(\xi) = \psi_0^{(0)} = \text{const} \quad (16)$$

3 Derivation of the Normalization Integral

3.1 5D Kinetic Term

The kinetic term for the graviton in the 5D action is:

$$S_{\text{kin}}^{(5)} = \frac{1}{2\kappa_5^2} \int d^5x \sqrt{-g^{(5)}} \left(-\frac{1}{4} g^{AC} g^{BD} \partial_A h_{CD} \partial_B h_{CD} + \dots \right) \quad (17)$$

where we show only the leading kinetic structure.

For the TT perturbation in the warped background, the 5D metric determinant gives:

$$\sqrt{-g^{(5)}} = e^{4A(\xi)} \quad (18)$$

(The exponent $4A$ arises from $\det(e^{2A}\eta_{\mu\nu}) = e^{8A}$ for the 4D block, combined with the $d\xi^2$ term; this follows from our metric convention (1).)

3.2 Reduction to 4D

Substituting the KK decomposition (9) with the zero-mode $\psi_0 = \text{const}$:

$$S_{\text{kin}}^{(5)} \supset \frac{1}{2\kappa_5^2} \int d^4x \int d\xi e^{4A} \left(-\frac{1}{4} |\psi_0|^2 \eta^{\mu\rho} \eta^{\nu\sigma} \partial_\mu h_{\rho\sigma}^{(0)} \partial_\nu h_{\rho\sigma}^{(0)} \right) \quad (19)$$

Factoring out the 4D integral:

$$S_{\text{kin}}^{(5)} \supset \frac{|\psi_0|^2}{2\kappa_5^2} \left(\int d\xi e^{4A(\xi)} \right) \int d^4x \left(-\frac{1}{4} \partial_\mu h_{\rho\sigma}^{(0)} \partial^\mu h^{(0)\rho\sigma} \right) \quad (20)$$

3.3 Identification of the 4D Planck Mass

The 4D Einstein-Hilbert action has the form:

$$S_{\text{EH}}^{(4)} = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g^{(4)}} R^{(4)} \quad (21)$$

At the linearized level, this gives a kinetic term:

$$S_{\text{kin}}^{(4)} = \frac{M_{\text{Pl}}^2}{2} \int d^4x \left(-\frac{1}{4} \partial_\mu h_{\rho\sigma} \partial^\mu h^{\rho\sigma} \right) \quad (22)$$

Matching (20) to (22):

$$\frac{M_{\text{Pl}}^2}{2} = \frac{|\psi_0|^2}{2\kappa_5^2} \int d\xi e^{4A(\xi)} \quad (23)$$

Using $\kappa_5^2 = 8\pi/M_5^3$:

$$M_{\text{Pl}}^2 = \frac{M_5^3}{8\pi} |\psi_0|^2 \int d\xi e^{4A(\xi)} \quad (24)$$

3.4 Normalization Convention

We adopt the normalization:

$$\int d\xi e^{4A(\xi)} |\psi_0(\xi)|^2 = 8\pi \mathcal{I} \quad (25)$$

where $\mathcal{I}$ has dimension of length.

Substituting into (24):

$$\boxed{M_{\text{Pl}}^2 = M_5^3 \mathcal{I}} \quad (26)$$

This is the **bridge relation** connecting the 4D and 5D Planck masses. (tag: [D])

4 Compact Flat Case: Explicit Evaluation of $\mathcal{I}$

4.1 Setup: Flat Extra Dimension

Consider the simplest case: a flat extra dimension with $A(\xi) = 0$ and compact topology $\xi \in [0, R_\xi]$ (or equivalently S^1 of circumference R_ξ).

The zero-mode equation (13) becomes:

$$\psi_0'' = 0 \quad (27)$$

The general solution is:

$$\psi_0(\xi) = a + b\xi \quad (28)$$

4.2 Boundary Conditions

For a compact dimension with periodic boundary conditions:

$$\psi_0(0) = \psi_0(R_\xi), \quad \psi_0'(0) = \psi_0'(R_\xi) \quad (29)$$

From (28):

$$a = a + bR_\xi \quad \Rightarrow \quad b = 0 \quad (30)$$

Therefore:

$$\psi_0(\xi) = a = \text{const} \quad (31)$$

4.3 Normalization

The normalization integral (25) with $A = 0$ becomes:

$$\int_0^{R_\xi} d\xi |\psi_0|^2 = |a|^2 R_\xi = 8\pi\mathcal{I} \quad (32)$$

We can choose:

$$|a|^2 = \frac{8\pi}{R_\xi} \quad (33)$$

which gives:

$$\mathcal{I} = \frac{1}{R_\xi} \cdot R_\xi = 1 \quad (\text{dimensionless convention}) \quad (34)$$

Alternatively, with $|a|^2 = 8\pi$:

$$\boxed{\mathcal{I} = R_\xi} \quad (35)$$

We adopt the convention (35) where $\mathcal{I}$ carries dimension of length. (tag: [D], conditional on flat compact model)

4.4 Physical Interpretation

The result $\mathcal{I} = R_\xi$ has a clear physical meaning: the effective 4D gravity strength is diluted by the volume of the extra dimension. A larger R_ξ means more “spreading” of gravity into the bulk, reducing the apparent 4D Planck mass.

5 Newton's Constant: Explicit Bridge

5.1 General Relation

Newton's gravitational constant in 4D is related to the Planck mass by:

$$G_N = \frac{1}{8\pi M_{\text{Pl}}^2} \quad (36)$$

Convention note: We use $M_{\text{Pl}} = 1.221 \times 10^{19}$ GeV throughout, which is the standard Planck mass $m_{\text{Pl}} = (\hbar c/G_N)^{1/2}$. The factor 8π in (36) follows from matching to the 4D Einstein-Hilbert action in the form $S = (M_{\text{Pl}}^2/2) \int \sqrt{-g} R$, which absorbs the $16\pi G_N$ denominator of the standard GR action.

Substituting the bridge relation (26):

$$G_N = \frac{1}{8\pi M_5^3 \mathcal{I}} \quad (37)$$

5.2 Compact Flat Case

For $\mathcal{I} = R_\xi$:

$$\boxed{G_N = \frac{1}{8\pi M_5^3 R_\xi}} \quad (38)$$

This is the **weak-field Newton law bridge** for a compact flat extra dimension. (tag: [D])

5.3 Gravitational Potential

The 4D gravitational potential between two masses m_1, m_2 separated by distance $r \gg R_\xi$ is:

$$V(r) = -\frac{G_N m_1 m_2}{r} = -\frac{m_1 m_2}{8\pi M_5^3 R_\xi \cdot r} \quad (39)$$

This recovers the standard $1/r$ behavior, confirming that the compact extra dimension is “invisible” at distances $r \gg R_\xi$.

5.4 Non-Compact Case (Remark)

For a non-compact extra dimension ($R_\xi \rightarrow \infty$), the integral $\mathcal{I}$ diverges. Regularization shows that the potential becomes:

$$V(r) \sim \frac{1}{r^2} \quad (\text{5D scaling}) \quad (40)$$

which violates the observed $1/r$ Newton law. This excludes non-compact, flat extra dimensions without warping. The compact or warped scenarios are required to recover 4D gravity.

6 Calibrated Closure

6.1 Solving for M_5

From the bridge relation (26) with $\mathcal{I} = R_\xi$:

$$M_{\text{Pl}}^2 = M_5^3 R_\xi \quad (41)$$

Solving for M_5 :

$$\boxed{M_5 = \left(\frac{M_{\text{Pl}}^2}{R_\xi} \right)^{1/3} = M_{\text{Pl}}^{2/3} R_\xi^{-1/3}} \quad (42)$$

This is the **calibrated closure formula**.

(tag: [D])

6.2 EW-Scale Identification

The compactification scale R_ξ is not derivable from EDC axioms (Track A NO-GO, see v16). We therefore identify it with an electroweak length scale:

$$\boxed{R_\xi = \frac{\hbar c}{M_Z^{\text{obs}}}} \quad (43)$$

with $M_Z^{\text{obs}} = 91.1876 \pm 0.0021$ GeV.

(tag: [I]+[BL])

This gives:

$$R_\xi = \frac{1.973 \times 10^{-16} \text{ GeV} \cdot \text{m}}{91.1876 \text{ GeV}} = 2.165 \times 10^{-18} \text{ m} \quad (44)$$

6.3 Baseline Inputs

Table 1: Baseline inputs for calibrated closure.

Quantity	Symbol	Value	Tag
4D Planck mass	$M_{\text{Pl}}^{\text{obs}}$	$1.221 \times 10^{19} \text{ GeV}$	[BL]
Z boson mass	M_Z^{obs}	$91.1876 \pm 0.0021 \text{ GeV}$	[BL]

6.4 Numerical Result

Substituting into (42):

$$M_5 = M_{\text{Pl}}^{2/3} M_Z^{1/3} = (1.221 \times 10^{19})^{2/3} \times (91.1876)^{1/3} \text{ GeV} \quad (45)$$

Computing:

$$M_{\text{Pl}}^{2/3} = (1.221 \times 10^{19})^{2/3} = 5.35 \times 10^{12} \text{ GeV} \quad (46)$$

$$M_Z^{1/3} = (91.1876)^{1/3} = 4.50 \text{ GeV}^{1/3} \quad (47)$$

Therefore:

$$\boxed{M_5 = 2.41 \times 10^{13} \text{ GeV}} \quad (48)$$

This is at the **GUT scale**, approximately $10^{-6} M_{\text{Pl}}$.

6.5 Error Budget

From error propagation:

$$\frac{\delta M_5}{M_5} = \sqrt{\left(\frac{2}{3} \frac{\delta M_{\text{Pl}}}{M_{\text{Pl}}}\right)^2 + \left(\frac{1}{3} \frac{\delta M_Z}{M_Z}\right)^2} \quad (49)$$

With $\delta M_{\text{Pl}}/M_{\text{Pl}} \approx 1.1 \times 10^{-5}$ and $\delta M_Z/M_Z \approx 2.3 \times 10^{-5}$:

$$\frac{2}{3} \frac{\delta M_{\text{Pl}}}{M_{\text{Pl}}} \approx 0.73 \times 10^{-5} \quad (50)$$

$$\frac{1}{3} \frac{\delta M_Z}{M_Z} \approx 0.77 \times 10^{-5} \quad (51)$$

Total:

$$\frac{\delta M_5}{M_5} \approx 1.1 \times 10^{-5} \approx 0.001\% \quad (52)$$

Table 2: Robustness of M_5 under different EW-scale identifications.

M_*	Value (GeV)	M_5 (GeV)	$\Delta \log_{10} M_5$	Tag
M_Z (canonical)	91.19	2.41×10^{13}	0 (ref)	[BL]
M_W	80.38	2.31×10^{13}	-0.018	[Dc]
v_{EW}	246.2	3.35×10^{13}	+0.143	[BL]

7 Robustness Under EW-Scale Choice

The identification (43) was tested with alternative EW scales (v17):

Verdict: All choices yield M_5 in the range $(2.3\text{--}3.4) \times 10^{13}$ GeV with $|\Delta \log_{10} M_5| < 0.15$. The closure is **order-of-magnitude stable**.

8 Epistemic Summary

Table 3: Epistemic ledger for the derivation.

Relation	Source	Tag
$M_{Pl}^2 = M_5^3 \mathcal{I}$	5D→4D reduction	[D]
$\mathcal{I} = R_\xi$ (flat compact)	Zero-mode normalization	[Dc]
$G_N = 1/(8\pi M_5^3 R_\xi)$	Bridge algebra	[D]
$M_5 = M_{Pl}^{2/3} R_\xi^{-1/3}$	Closure formula	[D]
$R_\xi = \hbar c/M_Z$	EW identification	[I]
$M_Z = 91.1876$ GeV	Experiment (PDG)	[BL]
$M_{Pl} = 1.221 \times 10^{19}$ GeV	Experiment (CODATA)	[BL]
$M_5 = 2.41 \times 10^{13}$ GeV	Numerical result	[D][BL]
R_ξ from axioms	Track A attempt	NO-GO

9 Conclusion

We have derived, from first principles:

1. The **bridge relation** $M_{Pl}^2 = M_5^3 \mathcal{I}$ from KK reduction of the 5D Einstein-Hilbert action.
2. The **normalization integral** $\mathcal{I} = R_\xi$ for a flat compact extra dimension.
3. The **Newton constant bridge** $G_N = 1/(8\pi M_5^3 R_\xi)$.
4. The **closure formula** $M_5 = M_{Pl}^{2/3} R_\xi^{-1/3}$.

Combined with the EW-scale identification $R_\xi = \hbar c/M_Z$ [I]+[BL], we obtain:

$$M_5 = 2.41 \times 10^{13} \text{ GeV} \pm 0.001\% \quad (53)$$

The gravity sector is **CLOSED (calibrated)** under minimal baseline inputs. Internal derivation of R_ξ remains **NO-GO**.